

John Perkins

Backend Software Engineer · AI/Distributed Systems

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EXPERIENCE

AI Engineer II

American Express (Contract)

Phoenix, AZ

February 2026 – Present

- **Agentic AI & Platform Engineering:**

- Designed and built the team's first end-to-end **agentic AI systems**: an **MCP server**, multiple LLM agents and workflows, and an **agent control plane** for orchestration.
- Shipped document-processing features end-to-end (filtering, audit logging, multi-format ingestion, **PII detection/redaction**) on a production platform processing **70k+ documents/day** of sensitive financial data.

Co-Founder & CTO

Notum

Phoenix, AZ

December 2025 - Present

- **AI Knowledge Platform for Law Firms:**

- Built a **ReAct retrieval agent** over a **Neo4j knowledge graph** and vector index, served via streaming **SSE**.
- Architected a **multi-tenant FastAPI backend** on **ECS Fargate** with **schema-per-tenant PostgreSQL** isolation.
- Engineered an **async ingestion pipeline (SQS + Lambda)** with hybrid OCR, processing **100k+ documents/run**.
- Owned security end-to-end: **WorkOS SSO**, VPC isolation, rate limiting, **OWASP ASVS Level 2** compliance.

Co-Founder

Scout

Phoenix, AZ

January 2025 - December 2025

- **Browser Automation Agent:**

- Built an **LLM agent** on a **LangGraph** plan-execute loop, driving **Playwright** from cleaned DOM snapshots.
- Cut repeat-site LLM calls by **90%** by replaying **cached action chains** keyed to page-state snapshots.
- Exposed a **FastAPI** job API with concurrency-capped workers, **human-in-the-loop** recovery, and **PostgreSQL** state.

Graduate Researcher

School of Computing and Augmented Intelligence, ASU

Tempe, AZ

August 2023 - May 2025

- **Adversarial Robustness of Classifiers Using Implicit Neural Representations:**

- Engineered adversarial-attack modules for INR classifier pipelines, identifying and patching key vulnerabilities.
- Built a gradient-approximating **surrogate model** to craft image-space attacks and benchmark defenses.

Software Engineer (Intern)

Choice Hotels International

Tempe, AZ

May 2023 - December 2023

- **Cross-Platform Ticket Synchronization between ServiceNow and Jira:**

- Engineered a **bidirectional sync system** between ServiceNow and Jira, processing **300+ tickets/day** with webhook ingestion, payload normalization, and conflict resolution, cutting resolution time from **weeks to days**.

PROJECTS

Real-Time Chat Service

Go · NATS · MongoDB · Docker

- Built a horizontally scalable real-time chat backend using **Go WebSockets** and **NATS pub/sub**, supporting 1:1 and group conversations with JWT authentication and achieving **5,000 messages/second at 99.9% delivery rate**.
- Architected stateless backend nodes with **MongoDB** persistence and **Chi router**, enabling seamless horizontal scaling for concurrent user sessions.
- Implemented WebSocket protocol with real-time typing indicators, read receipts, and cursor-based message pagination APIs.

URL Shortener

Go · gRPC · AWS · Redis · Docker · PostgreSQL

- Designed a high-throughput URL shortener with **gRPC**, **Redis** cache, and **PostgreSQL** storage; containerized with Docker.
- Load-tested to **5k req/s** at <150 ms p95 latency while maintaining **99.99% success rate**.
- Implemented CI/CD with **GitHub Actions** and **Terraform**, enabling one-click rollout to **AWS**.

SKILLS

Languages: Python, Go, C++, Java, Bash, SQL

Frameworks / Infra: FastAPI, gRPC, REST, WebSockets, Docker, Kubernetes, Terraform, GitHub Actions

Cloud / DevOps: AWS (ECS, ECR, SQS, RDS, ElastiCache, S3, Cognito, IAM), CI/CD, Git

Data / Reliability: Redis, PostgreSQL, MongoDB, Neo4j, Prometheus, Grafana, k6, Postman

EDUCATION

M.S. Computer Science - Arizona State University - 3.8 GPA

May 2025

B.S. Computer Science - Arizona State University - 3.47 GPA

Dec 2023